ENDOCRINE PHARMACOLOGY

TREATMENT OF DIABETES WITH INSULIN

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Join Dr. Hliebov's Office Hours (via Zoom):
https://zoom.us/j/96858712961?pwd=UklWVU1ETmhjRUd3UlAzVXM4WkpsQT09

Meeting ID: 968 5871 2961
Passcode: 680201

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Learning Objectives

1. Understand the physiology of insulin. Review the synthesis, secretion, and regulation of insulin in the pancreas. List the main factors regulating insulin secretion.
2. Explain the physiological role of insulin in glucose metabolism. Discuss the effects of insulin on carbohydrate, protein, and fat metabolism. Contrast the actions of insulin and the counterregulatory hormones on the liver and muscle.
3. Identify various types of insulin (e.g., rapid-acting, short-acting, intermediate-acting, long-acting, and ultra-long-acting). Compare and contrast the pharmacokinetics and pharmacodynamics of different insulin types. Be able to classify different types of insulin.
4. Explain the mechanism of action of insulin and glucagon at the cellular level. Discuss insulin receptors and the signaling pathways involved in insulin action.
5. Describe the clinical indications for insulin therapy, including Type 1 and Type 2 diabetes mellitus. Explain the choice of an insulin regimen in diabetics.
6. Compare different insulin therapy regimens (e.g., basal-bolus, split-mixed, continuous subcutaneous insulin infusion). Discuss the factors influencing the choice of insulin regimen for individual patients.
7. Discuss the role of insulin therapy in the management of diabetic ketoacidosis and hyperosmolar hyperglycemic state.
8. List the main events requiring an increase in insulin dosage in the diabetic patient during illness, surgery, and other special situations.
9. Explain the methods of insulin administration (e.g., syringes, pens, pumps). List the main factors affecting insulin absorption.
10. Identify common and rare side effects of insulin therapy and glucagon. Describe the techniques for self-monitoring of blood glucose and continuous glucose monitoring.

Learning Objectives

11. Discuss the management of hypoglycemia and other complications related to insulin therapy.
12. Discuss considerations for insulin therapy in special populations such as children, pregnant women, and the elderly.
13. Explain the concept of insulin resistance and its role in metabolic syndrome. Discuss the implications of insulin resistance for cardiovascular health and metabolic diseases.
14. Review recent advances in insulin formulations and delivery systems. Discuss emerging therapies and future directions in insulin therapy.

Type 1 Diabetes

Diabetes Mellitus Drugs: Drug Classes and Drugs to Know			
Note: all insulin preparations are either native insulin or very similar analogs that have nearly identical actions at insulin receptors.			
Short-acting insulins		**Long-acting insulin**	
insulin lispro	insulin regular	insulin NPH	insulin glargine

Type 2 Diabetes

Type 2 Diabetes Mellitus Drugs: Drug Classes and Drugs to know		
Biguanides metformin	**Sulfonylureas** glimepiride glyburide glipizide	**Thiazolidinediones** rosiglitazone
Alpha-glucosidase inhibitors acarbose	**Incretin mimetics** exenatide liraglutide	**Gliflozins (SGLT2 inhibitors)** canagliflozin
Meglitinides Repaglinide	**Dipeptidyl-Peptidase-4 (DPP-4) Inhibitors** Sitagliptin Linagliptin	**Dopamine-2 Agonist** Bromocriptine

Diabetes Mellitus

- Diabetes Mellitus (DM) is a heterogeneous group of disorders characterized by elevated blood sugar levels caused by an absolute or relative lack of insulin.
- One of the leading causes of death in the US: 75% of those afflicted with diabetes die from CHD.

Diabetes Mellitus

Type I (IDDM)

- Beta-cell failure at the outset – absolute lack of insulin.
- Insulin-dependent DM (IDDM).
- Destruction of islet cells is related to autoimmunity and environmental factors:
  - Environmental factors include certain drugs, chemicals, and viruses common in <30 years of age but can occur at any age.

Type 2

- Gradual beta-cell deterioration.
- More common (90%), patients have a strong genetic predisposition.
- They may have adequate insulin but are unable to be taken up by cells or have inadequate levels - **insulin resistance is an important factor.**
- It affects people primarily after age 40, and most of them are obese.
- Diet, exercise, and oral hypoglycemic agents for the early stages of disease.
- For late-stage disease, insulin therapy is necessary.

INSULIN THERAPY

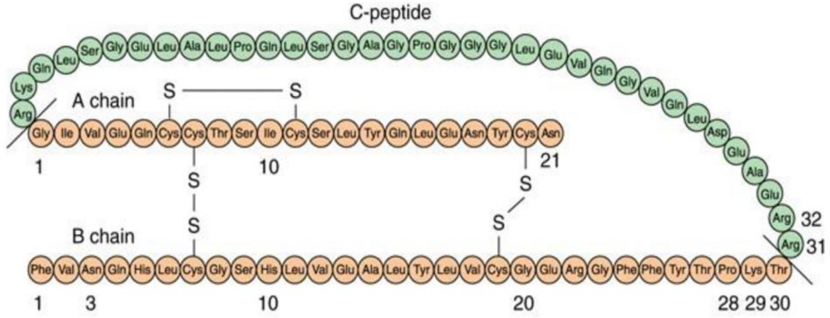
Insulin

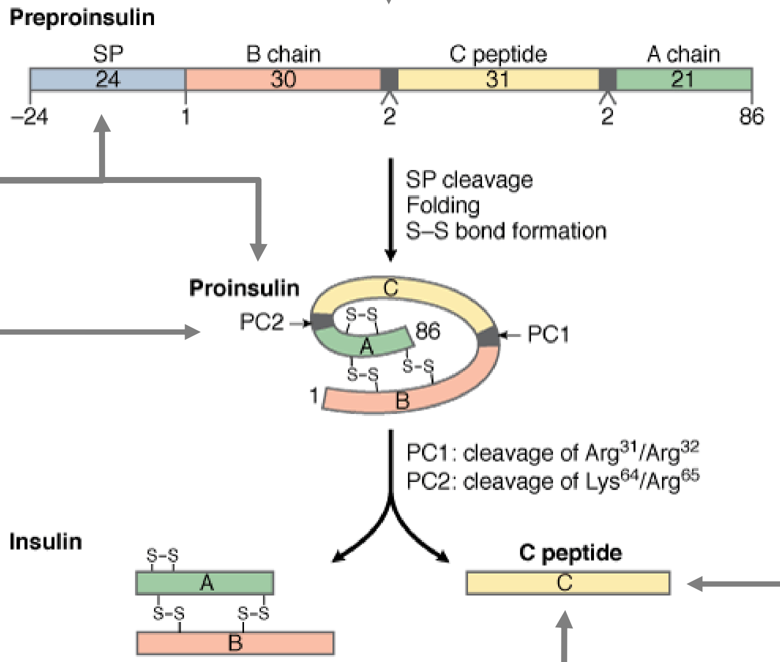
Synthesized in ß-cells of pancreatic islets as a single chain peptide preproinsulin (110 AA) from which 24 AAs are first removed to produce proinsulin.

Proinsulin results from cleavage of preproinsulin by removal of N-terminal signal peptide (SP) by peptidase.

Proinsulin is converted to insulin and C peptide:

- Insulin is a polypeptide hormone consists of two chains A and B, linked by 2 disulfide bridges.
- Contains 51 amino acids (A chain- 21 and B-chain 30 amino acids).
- Insulin is orally not active as degraded in the GIT.

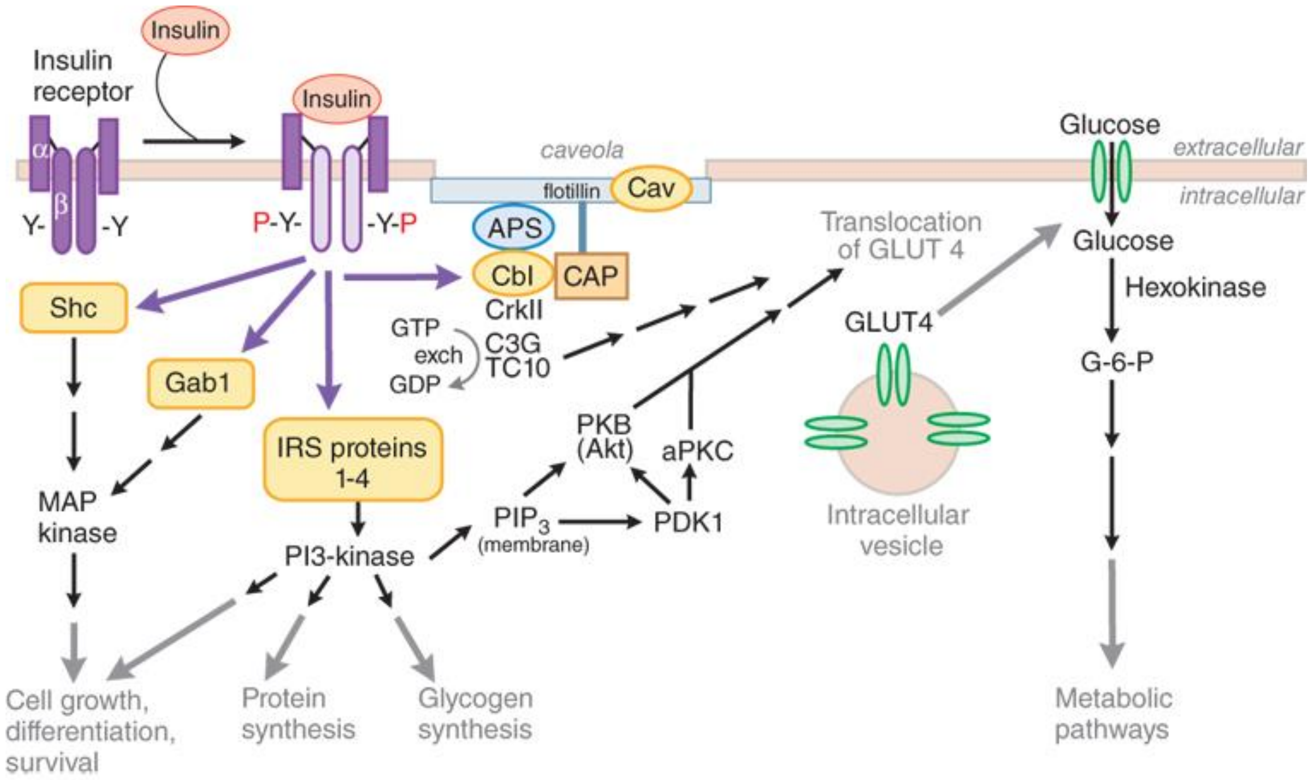




C-peptide is useful as a marker of endogenous beta-cell insulin secretion

Insulin: Receptors and Mechanism of Action

- **Tyrosine kinase receptor.**
- It is composed of two α subunits and two ß subunits.
- The α chains are entirely extracellular and house insulin-binding domains, while the linked beta chains penetrate through the plasma membrane.



Insulin: Mechanism of Action

Insulin binds to the alpha subunit of the insulin receptor

↓

Phosphorylation of tyrosine in beta subunit

↓

Tyrosine kinase activation (liver, adipose tissue, skeletal muscle*)

↓

Activate insulin receptor substrates (IRS, Gab1, Shc)

↓

IRS proteins

↓

Phosphatidylinositol-3-kinase (PI3-K) pathway

- Translocation of GLUT-1 and GLUT-4 glucose transporters → Increases glucose uptake into the cells (skeletal muscles & adipose tissue)
- Alters the phosphorylation state of key metabolic enzymes (enzymatic activation or inactivation) → Increase glycogen synthesis (stimulates glycogen synthase activity), alter protein and fat metabolism
- Induces the transcription of genes involved in glucose catabolism (e.g., glucose hexokinases). Inhibits explicitly transcription of other genes involved in gluconeogenesis.
- Stimulates Na^+/K^+-ATPase and promotes systemic cellular K^+ uptake

↓

Shc and Gab1

↓

MAP-kinase pathway

↓

Regulation of proliferation and differentiation of several cell types & regulation of DNA synthesis.

Actions of Insulin

Actions of Insulin

Liver	Skeletal Muscles	Adipose Tissue
• Inhibits gluconeogenesis (glucose production from protein, pyruvate, FFA, and glycerol) and increases glycolysis. • Inhibits glycogenolysis and stimulates glycogen synthesis (glycogenesis). • Increases the synthesis of triglycerides. • Increases protein synthesis.	• Increases glucose transport and glycolysis. • Increases glycogen deposition. • Increases protein synthesis.	• Increases glucose transport. • Increases triglyceride synthesis (lipogenesis). • Decreases intracellular lipolysis.

Overall effects of insulin: to favor storage of fuel. **Insulin promotes cellular K^+ uptake** and synthesis of proteins and fats (anabolic). Without insulin, most body cells cannot take up glucose.

Insulin Therapy

Preparations:

- All available preparations are either human insulin (produced by recombinant DNA techniques) or human analog insulin (some amino acids in the molecule are substituted or changed in position).
- Insulin is administered either **IV, IM, SC,** or inhaled (2014).

The duration of action of insulin can vary by:

- Modification of the insulin molecule (by recombinant technology).
- Conjugation of insulin with protamine in a low soluble complex. After injection, proteolytic enzymes degrade protamine, thus allowing slow absorption of insulin.
- Combination of insulin with zinc to form zinc salts. After injection, the salt precipitates, and insulin is released slowly.

Insulin Preparations

Rapid-acting – onset within 15 minutes and duration is < 5 hours; **only given SC.**
- Insulin Lispro
- Insulin Aspart
- Insulin Glulisine

Short-acting – duration of effect 5-8 hours; onset is 30 mins.
- Regular insulin (Crystalline Zinc Insulin/Soluble Insulin)
- Semilente Insulin

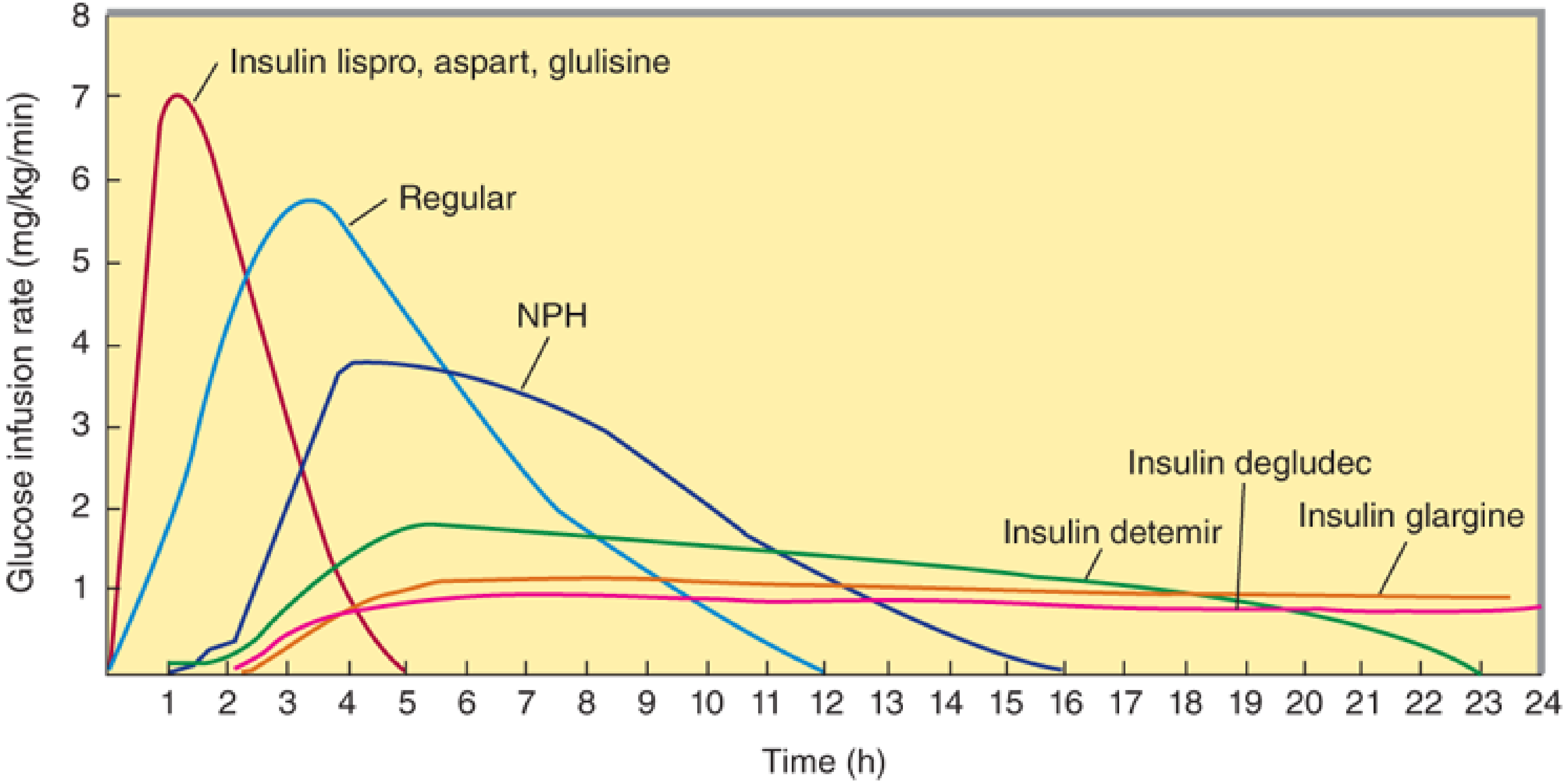
Intermediate-acting – duration of effect is around 24 hours; onset is a few hours, **only SC**.
- NPH (Neutral Protamine Hagedorn, Isophate)
- Lente (Ultra : Semi 7:3) insulin

Long-acting (slow-acting) – **only SC**; duration of effect around 30 hours.
- Ultralente
- Insulin Galargine
- Insulin Detemir
- Protamine-zinc insulin
- Insulin Degludec

Insulin Preparations (based on a depot, SC injection)			
Preparation	**Onset of action**	**Peak**	**Duration of action**
Rapid-acting			
Insulin lispro	< 15 min	30-60 min	2-4 hrs
Short-acting			
Insulin regular	~30 min	2-3 hours	3-6 hrs
Intermediate-acting			
Insulin NPH (Insulin isophane)	2-4 hrs	4-12 hrs	12-18 hrs
Long-acting			
Insulin glargine	1-2 hrs	No peak	20-24 hrs

Data from American Diabetes Association (note: these are general values only - you will find slight variations from different sources) (http://www.diabetes.org/living-with-diabetes/treatment-and-care/medication/insulin/insulin-basics.html

Insulin Preparations



Source: Todd W. Vanderah:
Basic & Clinical Pharmacology, Sixteenth Edition

Extent and duration of action of various types of insulin as indicated by the glucose infusion rates (mg/kg/min) required to maintain a constant glucose concentration. The durations of action shown are typical of an average dose of 0.2–0.3 U/kg. The duration of regular and NPH insulin increases considerably when the dosage is increased.

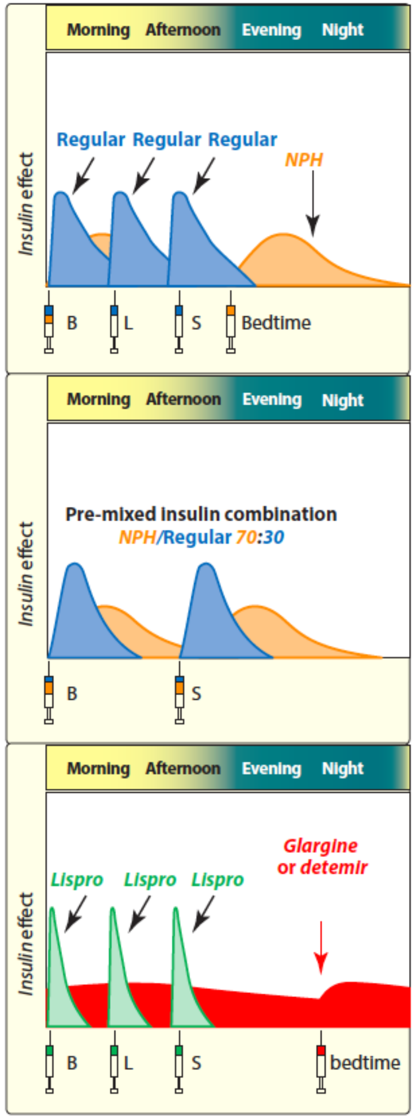
Therapeutic Uses of Insulin

- **Diabetes mellitus** (Type 1 and sometimes Type 2 if not well controlled).
- **Diabetic ketoacidosis** (diabetic coma).
- **Hyperosmolar** (nonketotic hyperglycemic) coma.

Increase in Dose Required:

- Infections
- Acute poisoning
- High fever
- Trauma, surgical operations
- Certain neoplasms
- Myocardial infarction
- Serious psychological stress
- Pregnancy
- Hyperthyroidism
- Diabetic ketoacidosis
- Use of drugs that cause hyperglycemia

Split-Mixed Regimen



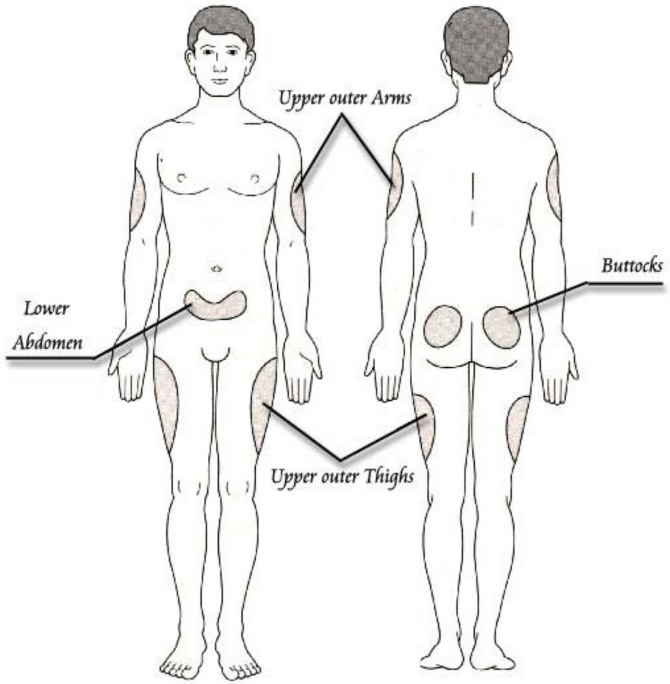
- Diabetic patients often need two types of insulin, **a basal long-acting/intermediate-acting insulin** and a **postprandial short-acting insulin,** as they have a very rapid onset of action with peak effects coinciding with peak postprandial hyperglycemia.

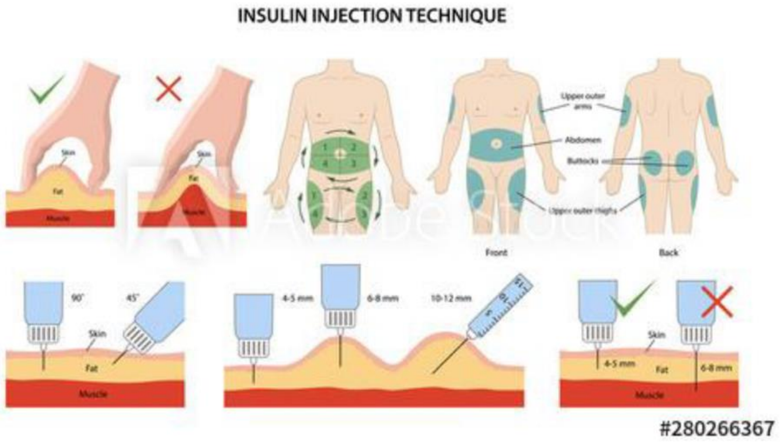
- A calculated daily dose of insulin is split into 2 or 3 doses, and insulin preparations used are of mixed types (rapid-acting+ intermediate-acting (30:70, 50:50, etc.) or rapid-acting + longer-acting, etc.).

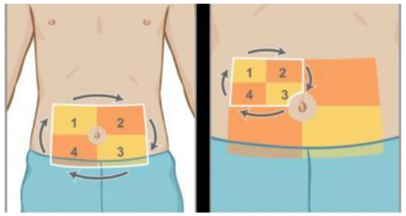
Examples of three regimens that provide both prandial and basal insulin replacement. B = breakfast; L = lunch; S = supper. NPH = neutral protamine Hagedorn.

Factors Affecting Insulin Absorption

- Site of injection.
- Type of insulin.
- Subcutaneous blood flow.
- Smoking.
- Muscle activity at the site of injection.
- Volume and concentration of insulin.
- Depth of injection.







Adverse Effects of Insulin

Hypoglycemia:

- The most frequent and potentially the most serious reaction.
- This can occur in any diabetic following inadvertent large doses of insulin injection, by missing a meal, or by performing a vigorous exercise.
- Symptoms can be related to:
  - (a) **counterregulatory sympathetic stimulation** sweating, anxiety, palpitation, tremor.
  - (b) **neuroglycopenia symptoms** (due to deprivation of the brain of its essential nutrient glucose) include dizziness, headache, behavioral changes, visual disturbances, fatigue, weakness, muscular incoordination, etc. --> ultimately, coma.

Local reactions:

- Swelling, erythema, and stinging sensation:
  Lipodystrophy (lipoatrophy or lipohypertrophy) of subcutaneous fat at the injection site is less likely with new preparations.

Allergic reactions (Anaphylactoid reactions):

- Due to contaminating proteins added to insulin in its formulation (protamine, zinc, etc.) or because of the sensitivity of one of the components (very rare with human/highly purified insulins).
- Local or systemic urticaria results from histamine release from tissue mast cells.

Weight gain (insulin is anabolic), **edema**, and **hypokalemia** (K^+ uptake into cells).

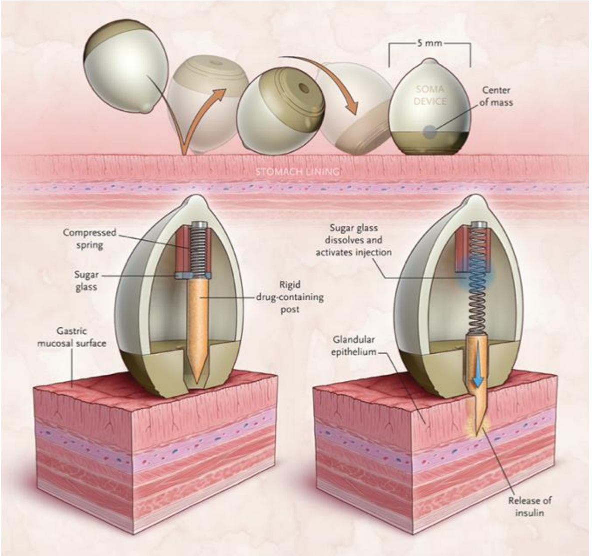
Insulin: Drug Interactions

- **Beta-blockers** prolong hypoglycemia by inhibiting compensatory mechanisms operating through ß2 receptors (ß 1 selective are less liable). Additionally, warning signs of hypoglycemia, like palpitation, sweating, tremor, and anxiety, are masked.
- **Thiazides, Furosemide, Corticosteroids, OCPs, etc.,** tend to raise blood sugar and reduce the effectiveness of insulin.
- **Acute ingestion of alcohol** can precipitate hypoglycemia by depleting hepatic glycogen.

Delivery of Insulin

- Disposable syringe (subcutaneous administration).
- Portable pen device.
- Insulin pump / continuous subcutaneous insulin infusion devices (CSII).
- Continuous glucose monitoring systems (e.g., Dexcom G7).
- **Bionic pancreas (e.g., iLet).**
- **Automatic insulin dosing systems (AID).**

- **Inhalation "Exubera"** – has led to nasal polyps, concern for lung cancers - **withdrawn from the market.**
- Recently was reported a preclinical test of a self-orienting millimeter-scale applicator (SOMA), which was designed to land on the gastric mucosal surface under the force of gravity and to orient in apposition to the plane of the tissue (Abramson, A., Caffarel-Salvador, E., Khang, M., et al. (2019) 'An ingestible self-orienting system for oral delivery of macromolecules', *Science,* 363, pp. 611-615).



Recall of MiniMed Insulin Pumps

U.S. FOOD & DRUG
ADMINISTRATION

MedWatch - The FDA Safety Information and Adverse Event Reporting Program

A MedWatch Safety Alert was just added to the FDA Medical Device Recalls webpage.

TOPIC: MiniMed Insulin Pumps by Medtronic: Class I Recall - Due to Potential Cybersecurity Risks

AUDIENCE: Patient, Endocrinology, Health Professional, Risk Manager

BACKGROUND: People who have diabetes may use the MiniMed insulin pump to deliver insulin for the management of their diabetes. The pump includes a remote controller which is designed to communicate wirelessly with the pump to deliver a specific amount of insulin to the person with diabetes.

ISSUE: Medtronic is recalling the specified insulin pumps due to potential cybersecurity risks. An unauthorized person (someone other than a patient, patient caregiver, or health care provider) could potentially record and replay the wireless communication between the remote and the MiniMed insulin pump. This person could instruct the pump to either over-deliver insulin to a patient, leading to low blood sugar (hypoglycemia), or stop insulin delivery, leading to high blood sugar and diabetic ketoacidosis, even death.

To date, the FDA is not aware of any reports of patient harm related to these potential cybersecurity risks.

iLet + iCGM

An official website of the United States government Here's how you know

U.S. FOOD & DRUG ADMINISTRATION

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FDA NEWS RELEASE

FDA Clears New Insulin Pump and Algorithm-Based Software to Support Enhanced Automatic Insulin Delivery

Agency Continues to Support Innovation of Next-Generation Technology in Diabetes Management

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For Immediate Release: May 19, 2023

Español

Today, the U.S. Food and Drug Administration cleared the Beta Bionics iLet ACE Pump and the iLet Dosing Decision Software for people six years of age and older with type 1 diabetes. These two devices, along with a compatible FDA-cleared integrated continuous glucose monitor (iCGM), will form a new system called the iLet Bionic Pancreas. This new automated insulin dosing (AID) system uses an algorithm to determine and command insulin delivery.

"Today's action will provide the type 1 diabetes community with additional options and flexibilities for diabetes management and may help to broaden the reach of AID technology," said Jeff Shuren, M.D., J.D., director of the FDA's Center for Devices and Radiological Health. "The FDA is committed to advancing new device innovation that can improve the health and quality of life for people living with chronic diseases that require day-to-day maintenance like diabetes through precision medicine approaches."

Questions?

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